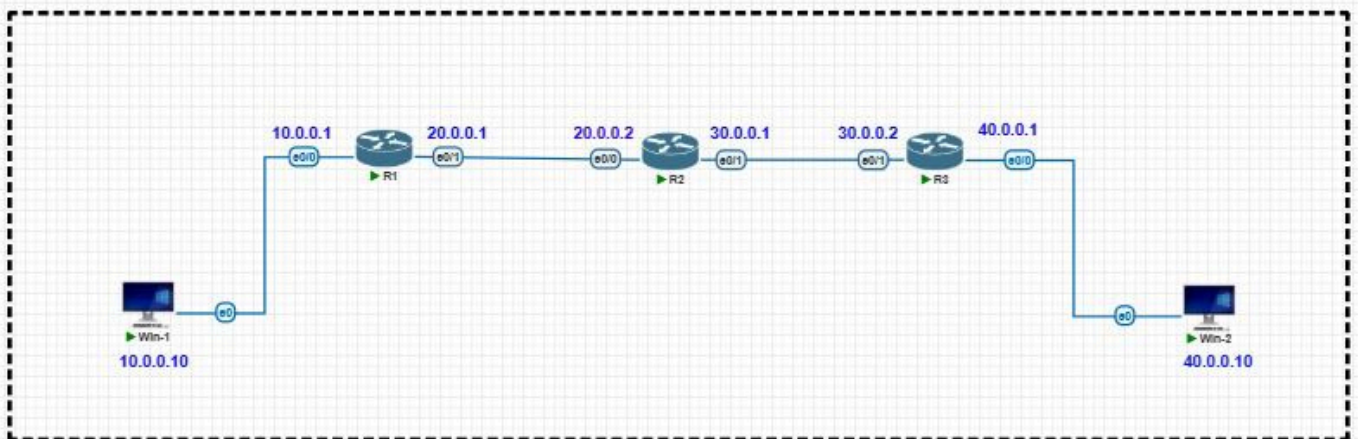




OSPF Router with ID (5.5.5.5) (Process ID 1)

Base Topology (MTID 0)

Topology priority is 64

Router is not originating router-LSAs with maximum metric

Number of areas transit capable is 0

Initial SPF schedule delay 5000 msec

Minimum hold time between two consecutive SPF's 10000 msec

Maximum wait time between two consecutive SPF's 10000 msec

Area 1

SPF algorithm last executed 00:00:50.357 ago

SPF algorithm executed 10 times

Area ranges are

R1(config)#do sh ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
70.0.0.10	1	FULL/BDR	00:00:36	20.0.0.2	Ethernet0/1

R1(config)#do sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.0.0.0/8 is directly connected, Ethernet0/0

L 10.0.0.1/32 is directly connected, Ethernet0/0

20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 20.0.0.0/8 is directly connected, Ethernet0/1

L 20.0.0.1/32 is directly connected, Ethernet0/1

O 30.0.0.0/8 [110/20] via 20.0.0.2, 00:02:18, Ethernet0/1

O 40.0.0.0/8 [110/30] via 20.0.0.2, 00:01:17, Ethernet0/1

R1(config)#

```
R2
R2(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
40.0.0.1         1     FULL/BDR        00:00:38    30.0.0.2     Ethernet0/1
5.5.5.5          1     FULL/DR         00:00:33    20.0.0.1     Ethernet0/0
R2(config)#
R2(config)#
R2(config)#
R2(config)#
R2(config)#do sh ip ospf neighbor
      ^
% Invalid input detected at '^' marker.

R2(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
40.0.0.1         1     FULL/BDR        00:00:31    30.0.0.2     Ethernet0/1
5.5.5.5          1     FULL/DR         00:00:37    20.0.0.1     Ethernet0/0
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O    10.0.0.0/8 [110/20] via 20.0.0.1, 00:05:27, Ethernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
O    40.0.0.0/8 [110/20] via 30.0.0.2, 00:04:11, Ethernet0/1
    70.0.0.0/32 is subnetted, 1 subnets
C    70.0.0.10 is directly connected, Loopback0
R2(config)#
```



```

R3
R3(config-router)#network 40.0.0.0 0.255.255.255 area 1
R3(config-router)#no sh
R3(config-router)#ex
R3(config)#
*Aug 19 15:57:57.718: %OSPF-5-ADJCHG: Process 3, Nbr 70.0.0.10 on Ethernet0/1 fr
om LOADING to FULL, Loading Done
R3(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
70.0.0.10        1     FULL/DR         00:00:38    30.0.0.1     Ethernet0/1
R3(config)#
*Aug 19 16:00:34.850: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R3(config)#
R3(config)#
R3(config)#
R3(config)#do sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
70.0.0.10        1     FULL/DR         00:00:36    30.0.0.1     Ethernet0/1
R3(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O        10.0.0.0/8 [110/30] via 30.0.0.1, 00:04:47, Ethernet0/1
O        20.0.0.0/8 [110/20] via 30.0.0.1, 00:04:47, Ethernet0/1
         30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C         30.0.0.0/8 is directly connected, Ethernet0/1
L         30.0.0.2/32 is directly connected, Ethernet0/1
         40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C         40.0.0.0/8 is directly connected, Ethernet0/0
L         40.0.0.1/32 is directly connected, Ethernet0/0
R3(config)#

```

